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System and method for MEMS devices

Abstract

Systems and methods for MEMS devices are disclosed. A microelectromechanical system (MEMS) device includes a substrate, and a MEMS structure supported by the substrate. The MEMS structure includes a first layer of a first material comprising a titanium aluminum alloy. The MEMS structure further includes an aluminum layer on the first layer.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to microelectromechanical systems (MEMS) and, more particularly, to system and method for MEMS devices.

BACKGROUND

(2) Microelectromechanical systems (MEMS) include microscopic devices that often include moving parts controlled through electrical signals. A digital micromirror device (DMD) is a particular example of a MEMS device that includes an array of micromirror assemblies that each include a mirror that can be rotated to direct the reflection of light on the mirror surface. An array of such micromirror assemblies may be fabricated on a single chip for implementation in a projector with each micromirror assembly controlling a separate pixel of a projected image.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an example DMD chip constructed in accordance with teachings disclosed herein.

(2) FIG. **2** is a cross-sectional view of an example micromirror assembly of the example DMD chip of FIG. **1** including a metal plate.

(3) FIG. **3** is an exploded view of the example micromirror assembly of FIG. **2**.

(4) FIG. **4** is a cross-sectional view of another example micromirror assembly.

(5) FIG. **5** is a cross-sectional view of another example micromirror assembly.

(6) FIG. **6** illustrates the example micromirror assembly of FIG. **2** with the metal plate curved downward.

(7) FIG. **7** illustrates the example micromirror assembly of FIG. **2** with the metal plate curved upward.

(8) FIGS. **8-12** illustrate different example micromirror structures that may be constructed to implement the micromirror structures in any one of FIGS. **2-5**.

(9) FIGS. **13** and **14** are charts showing measured values of the curvature of metal plates of experimental micromirrors on multiple different dies fabricated on different wafers, where different ones of the wafers are associated with metal plates having different metal stack designs.

(10) FIG. **15** is a flowchart illustrating an example method of manufacturing a micromirror assembly in accordance with teachings disclosed herein.

(11) FIGS. **16-25** illustrate different stages of fabrication of the example micromirror assembly described in connection with the example method of manufacture set forth in FIG. **15**.
(12) In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale. Instead, the thickness of the layers and/or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular.
(13) As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.
(14) Notwithstanding the foregoing, in the case of a semiconductor device, “above” is not with reference to Earth, but instead is with reference to a bulk region of a base semiconductor substrate (e.g., a semiconductor wafer) on which components of an integrated circuit are formed. Specifically, as used herein, a first component of an integrated circuit is “above” a second component when the first component is farther away from the bulk region of the semiconductor substrate than the second component.
(15) As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.
(16) As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.
(17) Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name.
(18) As used herein, “approximately” and “about” refer to dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections. As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time+/−1 second.
(19) As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.
DETAILED DESCRIPTION

(20) The fabrication processes involved during the manufacture of microelectromechanical system (MEMS) devices can produce stress gradients in components of such devices that can affect the final released shape of the components (e.g., the final shape after all processing and removal of surrounding sacrificial materials). For example, the stress gradient in a micromirror of a digital micromirror device (DMD) can cause the surface of the mirror to deflect or curve, thereby resulting in a non-planar surface. Deflections in the surface of a micromirror are generally undesirable because they reduce the efficiency with which the mirror is able to reflect light in a controlled manner (e.g., they can reduce contrast and tilt angle control).
(21) The stress gradient and, by extension, the resulting shape of a component of a MEMS device is dependent on the materials used for the component and the deposition process(es) used to deposit the materials when fabricating the component. In many situations, particular materials for a component are needed to enable the component to function properly and/or to facilitate the fabrications processes involved such that using a different material is not a viable option to control or adjust the stress gradient and resulting shape of the component. Some fabrication processes that have been implemented to affect the stress gradient of micromirrors include an air-break in which a native oxide interlayer or film is allowed to form on a surface of one or more layers within the metal stack of the mirror substrate. While an air-break can be included in the fabrication process to affect the stress gradient, such an operation has a relatively limited impact on the final shape of MEMS components and can have deleterious impacts on the structural integrity of the components. Furthermore, the impact on the final shape of MEMS components cannot be precisely controlled.
(22) Examples disclosed herein enable the relatively precise control of stress gradients in components, structures, and/or elements of MEMS devices to select and/or tune the stress gradients, thereby selecting or tuning the final shapes of the associated components. More particularly, examples disclosed herein employ one or more layers of titanium aluminide (TiAl.sub.3) in components that have a base substrate material of aluminum (Al). The number of the titanium aluminide layers, the thickness of the layers, and the placement of the layers can all be used to adjust and control the resulting stress gradient. Furthermore, precise control of the stress gradient is possible because deposition processes can be precisely controlled to place titanium aluminide at specific locations with specific thickness in a relatively consistent manner. Further still, unit processes (e.g., deposition processes) are easily measurable and controllable. Additionally, the capability of in-situ deposition of titanium aluminide results in a reduced (e.g., minimal) impact on the manufacturability of an overall metal stack.
(23) FIG. **1** illustrates an example DMD chip **100** constructed in accordance with teachings disclosed herein. As shown in the illustrated example, the DMD chip **100** includes an array **102** of individual micromirror assemblies **104**. Many DMD chips include hundreds of thousands of individual micromirror assemblies **104**. However, a fewer number of micromirror assemblies **104** are shown in FIG. **1** for purposes of illustration. Each micromirror assembly **104** includes a micromirror (or simply mirror, for short) that can be rotated about a corresponding hinge via associated control circuitry in the DMD chip **100**. Further detail regarding the implementation and construction of each of the micromirror assemblies **104** is provided below in connection with FIG. **2**.
(24) FIG. **2** is a cross-sectional view of an example micromirror assembly **104** of the example DMD chip **100** of FIG. **1** taken along the plane defined by the lines **2-2** in FIG. **3**, which is an exploded view of the example micromirror assembly of FIG. **2**. As shown in the illustrated example, the micromirror assembly **104** is provided on an underlying substrate **202**. In some examples, the substrate **202** is a semiconductor (e.g., silicon) substrate. In some examples, all of the micromirror assemblies **104** of the DMD chip **100** of FIG. **1** are fabricated simultaneously on a common substrate (e.g., a single silicon wafer). More particularly, in some examples, multiple DMD chips (each with a corresponding array of micromirror assemblies **104**) are fabricated on a common substrate during the same fabrication processes.

(25) As shown in FIGS. **2** and **3**, the top of the micromirror assembly **104** (e.g., the point farthest away from the substrate **202**) is a plate **204** with an exterior or exposed surface **206** that is reflective to serve as the mirror for the micromirror assembly **104**. Accordingly, the plate **204** is alternatively referred to herein as the mirror plate, the micromirror, or simply the mirror of the micromirror assembly **104**. In this example, the plate **204** has a generally rectangular or square shape (as shown in FIG. **3**) but may be shaped in any other suitable manner (e.g., circular, hexagonal, etc.). In some examples, the plate **204** includes and/or is manufactured from a stack of metal layers. In some examples, a base or primary metal used in the metal stack is aluminum. In some examples, an uppermost layer (e.g., the layer that is exposed at the exterior surface **206**) that serves as the reflective surface of the mirror is aluminum. In some examples, the metal stack include one or more layers of titanium aluminide with a thickness selected to adjust or control the stress gradient within the plate **204** and, therefore, the final shape of the plate **204**. Further detail regarding different example metal stacks for the plate **204** is provide below in connection with FIGS. **7-11**.
(26) In the illustrated example of FIGS. **2** and **3**, the plate **204** is suspended in free space with support of a post **208** that protrudes from a back side **210** of the plate **204** (e.g., opposite the exterior or exposed surface **206**) near a center of the plate **204**. As shown in FIG. **3**, the back side **210** is a second exposed surface of the plate **204** that faces in an opposite direction to the exposed surface **206** on the top side of the plate **204**. In some examples, the post **208** is coupled to the plate **204** at a location other than the center of the plate **204**. In some examples, the plate **204** is supported by more than one post **208**. In some examples (as shown), the post **208** is integrally formed with the plate **204**. That is, in some examples, the post **208** includes walls defined by metal that protrudes downward from and is a continuous extension of one or more layers in the metal stack of the plate **204**. As shown, the formation of the post **208** (not visible in FIG. **3**) results in a hole **209** in the exterior surface **206** of the plate **204** that corresponds to the inside of the post **208**. In other examples, the hole **209** can be filled with a filler material and/or covered as discussed in more detail below. For purposes of clarity, the plate **204** and the post **208** are collectively referred to herein as a micromirror structure **213** of the micromirror assembly **104**.
(27) In this example, the post **208** is coupled to a hinge assembly **212** that includes a hinge **214** and hinge tips **215** formed of a common layer of material. As shown in the illustrated example, the hinge **214** and the hinge tips **215** are supported spaced apart from a top surface **216** of the substrate **202** by plurality of pillars **218**. As represented in the illustrated example, the pillars **218** have a hollow interior. In other examples, the pillars **218** may be solid. The hinge **214** is composed of a flexible material to enable the movement of the plate **204** by deflection, twisting, or bending of the hinge **214**. More particularly, in the illustrated example, the hinge **214** is to twist or bend so that the plate **204** rotates about an axis extending along a longitudinal length of the hinge **214**. In this example, the hinge assembly **212** is supported by a hinge base plate **217** positioned on the underlying substrate **202**.
(28) In some examples, movement of the plate **204** is controlled by electrical signals provided to one or more electrodes **220** positioned adjacent the hinge assembly **212**. In the illustrated example, the electrodes are defined by pillars **222** and flanges **224** at an upper end of the pillars **222**. In some examples, the pillars **222** have a similar height as the pillars **218** such that the flanges **224** are at a same height as the hinge **214** and hinge tips **215**. As represented in the illustrated example, the pillars **222** have a hollow interior. In other examples, the pillars **222** may be solid. In this example, separate electrodes **220** are positioned on either side of the hinge **214** and are supported by separate electrode base plates **226** positioned on the underlying substrate **202**. Charges applied to the electrodes **220** either attract or repel the plate **204**, the post **208**, and/or portions of the hinge assembly **212**, thereby enabling the plate **204** to rotate or move due to deflection of the hinge **214**. In some examples, charges applied to the electrodes **220** are provided through circuitry **302** provided in the substrate **202** (the circuitry **302** is diagrammatically represented on the top surface

216 of the substrate **202** in FIG. **3** for purposes of explanation). The position and size of the electrodes **220** shown in FIG. **2** is for purposes of illustration only.

(29) The particular design of the example micromirror assembly **104** shown in FIGS. **2** and **3** is for purposes of illustration only and many other designs are possible. For instance, FIG. **4** illustrates a cross-sectional view of another example micromirror assembly **400** in which a micromirror structure **402** (including a plate **404** and a post **406**) is supported near an end of a cantilevered hinge **408** supported by a pillar **410**. In such examples, unlike the hinge **214** of FIGS. **2** and **4** that twists between two pillars **218** at opposite ends of the hinge **214**, the hinge **408** of FIG. **4** deflects up and down at its free end (farthest from the pillar **410**) to cause the plate **404** to rotate about an axis proximate the pillar and extending into and out of the view shown in FIG. **4**. In this example, an electrode **412** is positioned adjacent the free end of the hinge **408**. The electrode **412** includes a pillar **414** that supports a flange **416**. In the illustrated example of FIG. **4**, the pillar **410** and electrode **412** are mounted on the underlying substrate **418** without an intervening base plate (as in FIGS. **2** and **3**).

(30) FIG. **5** illustrates a cross-sectional view of another example micromirror assembly **500** in which a micromirror structure **502** (including a plate **504** and a post **506**) is positioned on a hinge **508** that extends into and out of the drawing between separate pillar (not shown) in a manner similar to that shown in FIG. **3**. In the illustrated example of FIG. **5**, the hinge **508** is shown deflected to one side resulting in the plate **504** tilting or rotating accordingly. Similar to FIGS. **2** and **3**, the micromirror assembly **500** of FIG. **5** includes electrodes **510** (defined by pillars **512** and corresponding flanges **514**) positioned on either side of the hinge **508**. However, unlike the example shown in FIGS. **2** and **3**, the electrodes **510** in FIG. **5** are spaced farther away from hinge **508** with a longer protruding portion of the flanges **514** on the side of the electrodes **510** opposite the hinge **508**. In some examples, the plate **504** contacts the protruding portion of the flanges **514** (coupled to the pillars **512** of the electrodes **510**) when the micromirror **502** is rotated as shown in FIG. **5**. Further, the micromirror assembly **500** of FIG. **5** differs from FIGS. **2** and **3** in that both electrodes **510** are supported by a common portion of a base plate **516** (on an underlying substrate **518**) rather than separate portions as in FIGS. **2** and **3**. More generally speaking, the design, shape, and/or structure of any of the micromirror assemblies **104**, **400**, **500** of FIGS. **1-5** can be modified in any suitable manner in accordance with teachings disclosed herein. For instance, the hinge can be sized and shaped in any suitable manner and located in any suitable manner to enable adjustments to the orientation of the mirror. In some examples, more than one hinge may be implemented to enable adjustments to the orientation of the mirror in multiple directions. In some such examples, different hinges and associated supporting pillars may be positioned at different levels in an associated hinge assembly (e.g., a first hinge can be supported by a first pillar that is itself mounted to a second hinge supported by a second pillar). Further, the electrodes can be sized and shaped in any suitable manner and located at any suitable position relative to the micromirror structure and supporting hinge assembly.

(31) For purposes of explanation, the illustrated example of FIGS. **2** and **3** will be references with the understanding that the following description can be applied equally to the examples shown in FIGS. **4** and **5** and/or any other suitable micromirror assemblies. In some examples, the exterior surface **206** (e.g., the mirror surface) of the plate **204** is constructed to be a near planar surface to facilitate controlled reflections of light. However, due to limitations in manufacturing processes and resulting stresses created in the plate **204** and other components during such processes, the exterior surface **206** of the plate **204** is unlikely to be perfectly planar. More particularly, the fabrication processes involved in manufacturing the plate **204** supported on the hinge assembly **212** can result in a stress gradient across the plate **204** that can result in the plate **204** deforming. Such deformation is particularly problematic due to the fact that the plate **204** is positioned so as to be suspended in free space (except for the point at which it is coupled to the post **208**) such that there are no surrounding structures attached to the plate **204** to reduce its deformation.

(32) The particular way in which the plate **204** deforms can depend upon the materials used in the metal stack of the plate **204**, the thickness of the layers of such materials, and the fabrication processes involved in fabricating the plate **204** as well as any fabrication processes implemented after the fabrication of the plate **204**. For instance, in some situations, the stress gradient in the plate **204** can cause the plate to curve downward, thereby forming a convex exterior surface **206** as shown in the illustrated example of FIG. **6**. In other situations, the stress gradient in the plate **204** can cause the plate to curve upward, thereby forming a concave exterior surface **206** as shown in the illustrated example of FIG. **7**.

(33) The amount of deflection of the plate **204** in FIGS. **6** and **7** is exaggerated for purposes of explanation. However, actual measurements of typical micromirrors have that are not constructed in accordance with teachings disclosed herein have shown deflections that vary across the surface of the mirror between approximately +1400 Å near the corners and approximately −900 Å near the center of the plate. Thus, the mirror surface is non-planar with a total amount of deflection or non-planar variability across the surface area of the mirror exceeding 2000 Å. By contrast, actual measurements of experimental micromirrors constructed in accordance with teachings disclosed herein have shown significantly smaller amounts of deflection across the surface area of the mirror. More particularly, in some examples, the amount of deflection or non-planar variable across example micromirrors disclosed herein is approximately 150 Å. As such, examples disclosed herein provide mirror surfaces that are much less warped and significantly more planar than is possible using existing fabrication techniques.

(34) The improved planarity of the mirror surface in disclosed examples is accomplished by using one or more layers of metal that are composed of an alloy containing both titanium and aluminum (e.g., titanium aluminide (TiAl.sub.3)) adjacent one or more layers of metal that are composed of substantially pure aluminum. As used herein, “substantially pure aluminum” is expressly defined to mean at least 95 atomic percentage (at %) of the material is pure aluminum. As used herein, an “aluminum layer” is similarly defined to mean a layer that contains at least 95 at % of pure aluminum. By contrast, as used herein, a “titanium aluminum alloy” is expressly defined to refer to a material in which there is a significant amount (e.g., more than trace amounts) of each of titanium and aluminum. As used herein, “a significant amount” is expressly defined to mean more than 5 at % of a particular material (e.g., titanium or aluminum) is included in the alloy. Thus, as used herein, “titanium aluminum alloy” means an alloy that includes more than 5 at % of titanium and more than 5 at % of aluminum. More particularly, in some examples, the titanium aluminum alloy corresponds to titanium aluminide (TiAl.sub.3) with the proportion of titanium in the titanium aluminide ranging from 23 at % to 52 at %. In view of the above definitions, it should be noted that, in some examples, the substantially pure aluminum may include some titanium but in quantities that are less than a significant amount as defined above. For instance, in some examples, the “substantially pure aluminum” (or “aluminum layer”) contains less than 0.5 at % titanium or less (e.g., 0.2 at %). FIGS. **8-12** illustrate different example micromirror structures **800**, **900**, **1000**, **1100**, **1200** that may be constructed to implement any one of the micromirror structures **213**, **402**, **502** of FIGS. **2-5**. As shown in the illustrated examples, the micromirror structures **800**, **900**, **1000**, **1100**, **1200** are composed of a metal stack that includes multiples layers of different metal materials stacked on one another in a direction normal to the exterior planar surface (e.g., the exterior surface **206**) of the metal plates associated with the micromirror structures **800**, **900**, **1000**, **1100**, **1200**. More particularly, in the illustrated examples, the different layers are stacked parallel to one another and parallel to the exterior surface. In the illustrated examples, there is at least two layers of material that include substantially pure aluminum without a significant amount of titanium and at least one layer of material that includes significant amounts of both titanium and aluminum (e.g., titanium aluminide (TiAl.sub.3)). Further, as shown in the illustrated example, at least some of the layers of metal in the metal stack that is used to form the plate or micromirror also extends down into and forms the walls of the post that is used to couple the micromirror structure **800**, **900**, **1000**,

1100, **1200** to a hinge (e.g., any one of the hinges **214**, **408**, **508**). That is, in some examples, the post is integrally formed with the plate.

(35) Turning in detail to the illustrated examples, the micromirror structure **800** of FIG. **8** includes three layers **802**, **804**, **806** of material that define the micromirror or plate **808** and also define the post **810**. In this example, the layer **802** is composed of titanium aluminide (e.g., it includes significant amounts of both titanium and aluminum) whereas the layers **804**, **806** are composed of substantially pure aluminum (e.g., the layers **804**, **806** do not include a significant amount of titanium). Thus, in this example, there is not a significant amount of titanium between the layers **804**, **806**. In this example, there is a single layer of material that includes a significant amount of titanium that is farthest from the exterior surface **812** and closest to the underlying substrate (e.g., the substrate **202** shown in FIG. **2**). As shown in FIG. **8**, the material of both the layers **802**, **804** extend along and define the walls of the post **810**. In some examples, the thicknesses of the layers **802**, **804** are insufficient to fill the entire cross-section of the post **810**. Accordingly, in some examples, a filler material **814** is deposited in a cavity within the post **810** beneath the layer **816** to fill the gap between the layers **802**, **804**. The filler material **814** can be any suitable material (e.g., a photoresist, an organic bottom antireflective coating (BARC), etc.). The layer **806** extends across the cavity within the post **810** to cover the entire surface of the plate. More particularly, in this example, the layer **806** defines the exposed exterior surface **812** that serves as the mirror surface to reflect light.

(36) In some examples, the plate **808** has a total thickness **818** in the range of approximately 1200 Å to approximately 4000 Å. Different thickness can be used for different ones of the layers **802**, **804**, **806** in different designs of the plate **808** to achieve different stress gradients within the plate **808**. Thus, the stress gradient can be controlled or tuned in a precise manner by controller the thicknesses of the each of the layers **802**, **804**, **806**. In the illustrated example, the uppermost layer of aluminum that provides the mirror surface (e.g., the layer **806** in FIG. **8**) has a thickness **820** that is thicker than the other layers **802**, **804** in the metal stack. However, in other examples, the layer **806** has a thickness **820** that is less than or equal to one or both of the other layers **802**, **804**. In some examples, the layer **806** has a thickness **820** that is between one quarter and one half (e.g., approximately one third) the total thickness **818** of the plate. That is, in some examples, the thickness **820** of the layer **806** ranges from approximately 300 Å to approximately 2000 Å. Further, in the illustrated example, the layer **804** (also composed of aluminum) has a thickness **822** that is thicker than the layer **802** (composed of titanium aluminide). However, in other examples, the layer **804** has a thickness **822** that is less than or equal to the layer **802**. More particularly, in some examples, the thickness **822** of the layer **804** ranges from approximately 300 Å to approximately 2000 Å. Particular example thicknesses **822** for the layer **804** are shown and described in connection with FIGS. **13** and **14**. In some examples, the layer **802** (composed of titanium aluminide) has a thickness **824** that is equal to or greater than one or both of the other layers **804**, **806** (composed of aluminum). In other examples, the thickness **824** of the layer **802** is less than both the other layers **804**, **806**. More particularly, in some examples, the thickness **824** of the layer **802** ranges from approximately 100 Å to approximately 500 Å. Particular example thicknesses **824** for the layer **802** are shown and described in connection with FIGS. **13** and **14**. In some examples, the combined thickness of the layers composed of substantially pure aluminum (e.g., the layers **804**, **806** in FIG. **8**) is greater than the thickness of the layer composed of an alloy of titanium and aluminum (e.g., titanium aluminide). Thus, in some examples, a majority of the thickness of the plate **808** is composed of substantially pure aluminum.

(37) The example micromirror structure **900** shown in FIG. **9** includes two layers of titanium aluminide. More particularly, as shown in FIG. **9**, a layer **902** is titanium aluminide. Also, the layer **904** in FIG. **9** is substantially pure aluminum. Further, the uppermost layer **906** in FIG. **9** is substantially pure aluminum. The micromirror structure **900** of FIG. **9** also includes a layer **908** of material positioned between the layers **904**, **906**. In this example, the material of the layer **908** is

the same material as the layer **902**. That is, the layer **908** includes titanium aluminide. Thus, in this example, there is a significant amount of titanium positioned between the two layers **904**, **906** containing substantially pure aluminum. In some examples, as represented in FIG. **9**, the layer **908** (second layer of titanium aluminide) has a thickness **910** that is thinner than the layer **902** (first layer of titanium aluminide). In other examples, the layers **902**, **908** have a similar thickness. In other examples, the layer **902** is thinner than the layer **908**. More particularly, in some examples, the thickness **910** of the layer **908** ranges from approximately 50 Å to approximately 500 Å (e.g., 100 Å). In some examples, the combined thickness of the layers composed of substantially pure aluminum (e.g., the layers **904**, **906**) is greater than the combined thickness of the layers composed of titanium and aluminum (e.g., the layers **902**, **908**). In some examples, the thickness of one or more of the layers **902**, **904**, **906** in FIG. **9** are adjusted relative to the thicknesses of the corresponding layers **802**, **804**, **806** in FIG. **8** so that the layer **908** can be included while maintaining the total thickness **912** of the plate **914** in FIG. **9** the same as the total thickness **818** of the plate **808** in FIG. **8**. In the illustrated example of FIG. **9**, the titanium aluminide extends continuously from the layers **902**, **904**, **908** along the wall of the post **916** in a similar manner to the layers **802**, **804** as described above in connection with FIG. **8**. In this example, the post **916** is filled with a filler material **918** in a similar manner to the filler material **814** of FIG. **8**. While the micromirror structure **900** of FIG. **9** includes two layers of substantially pure aluminum and two layers of titanium aluminide, in other examples, the metal stack may include additional layers of substantially pure aluminum and/or additional layers of titanium aluminide.

(38) In the example micromirror structure **1000** shown in FIG. **10** a layer **1002** includes substantially pure aluminum, a layer **1004** includes titanium aluminide, and an uppermost layer **1006** includes substantially pure aluminum. Thus, in this examples, the layer **1002** corresponds to the only layer of titanium aluminide in the micromirror structure **1000** and is positioned between the layers **1002**, **1006**. As described above, each of the layers **1002**, **1004**, **1006** may have any suitable thickness (such as those described for the corresponding layers in FIGS. **8** and **9**) to define a corresponding total thickness for the plate **1008** and the walls of the associated post **1010**. In this example, the post **1010** is filled with a filler material **1012** in a similar manner to the filler material **814**, **918** of FIGS. **8** and **9**.

(39) The example micromirror structure **1100** shown in FIG. **11** has a bottommost layer **1102** of titanium aluminide, layers **1104**, **1106** of substantially pure aluminum, and a layer **1108** of titanium aluminide between the two layers **1104**, **1106** of the aluminum. However, the upper layer of titanium aluminide (e.g., the layer **1108**) does not extend along a wall of or into the post **1110**. That is, the layer **1108** extends continuously across the plate from one edge to an opposing edge by crossing over the filler material **1112** inside the post **1110**. This arrangement is achieved by changing the order of operations implemented to fabricate the micromirror structure **1100** of FIG. **11** relative to the order of operations followed to fabricate the micromirror structure **900** of FIG. **9**. More particularly, to fabricate the example micromirror structure **1100** of FIG. **11**, the filler material **814** is added before the layer **1108** is deposited. By contrast, to fabricate the example micromirror structure **900** of FIG. **9**, the filler material **814** is added after the layer **908** is deposited.

(40) In each of FIGS. **8-11** each of the layers **802**, **804**, **806**, **902**, **904**, **906**, **908**, **1002**, **1004**, **1006**, **1102**, **1104**, **1106** in the metal stack extend substantially across the entire area of the example plates **808**, **914**, **1008**, **1114**. However, in some examples, as represented in FIG. **12**, at least one of the layers of titanium aluminide is limited to laterally isolated portions of the plane in which the layer is located. More particularly, the micromirror structure **1200** of FIG. **12** includes two layers **1202**, **1204** of titanium aluminide and two layers **1206**, **1208** of substantially pure aluminum. In this example, the upper layer **1204** of titanium aluminide does not extend a full way across the plate **1210**. Stated differently, whereas the aluminum layers **1206**, **1208** extend across the entire area of the plate **1114**, the layer **1204** of titanium aluminide extends across an area that is smaller than the areas associated with the aluminum layers **1206**, **1208**. More particularly, in the example shown in

FIG. **12**, the titanium aluminide in the upper layer **1204** of titanium is located in regions adjacent the outer edges or perimeter of the plate **1210** and spaced apart from the center of the plate **1210** and the post **1212**. However, any other suitable placement of the titanium aluminide is possible (e.g., near the center of the plate **1210** and spaced apart from the outer edges of the plate **1210**, only at the corners of the plate **1210**, etc.). The particular location of the titanium aluminide depends upon the particular structure of the plate **1210**, the thicknesses of the layers within the metal stack defining the plate **1210**, and the desired stress gradient and corresponding final shape of the plate **1210**.
(41) Placing titanium aluminide at particular locations, as represented in FIG. **12**, can have particular impacts on the stress gradient in the plate **1210**. Thus, by precisely controlling the location of the titanium aluminide, the stress gradient can be precisely controlled, thereby enabling the precise control of the final released shape of the plate **1210**. However, depositing titanium aluminide on limited regions rather than the entire surface of the underlying layer in the metal stack creates complexities in the fabrication process. Accordingly, in some examples, the titanium aluminide is deposited in layers that cover all or substantially all of the underlying layers in the metal stack as represented in FIGS. **8-11**. In such examples, relatively precise control of the stress gradient is still possible by selecting the number of titanium aluminide layers in the metal stack, the placement of the titanium aluminide layers in the metal stack (e.g., the order in which the layers are deposited to create the stack), and the thicknesses of the titanium aluminide layers (as well as the thickness of the aluminum layers) in the metal stack.
(42) The impact of different thicknesses and placements of the layers has been demonstrated through experimental testing. In particular, FIGS. **13** and **14** are charts showing measured values of the curvature of metal plates of micromirror structures in experimental DMDs on different dies fabricated on different wafers, where different ones of the wafers are associated with metal plates having different metal stack designs. More particularly, the first two wafers (W**1**, W**2**) represented on the lefthand side of the charts in both FIGS. **13** and **14** are the same and represent a reference or baseline. As shown in the charts, the first two wafers (W**1**, W**2**) include DMDs with metal plates (e.g., micromirror structures) having a first metal layer of titanium aluminide with a thickness of 100 Å, a second metal layer of substantially pure aluminum with a thickness of 775 Å, and a third metal layer of titanium aluminide with a thickness of 100 Å. Thus, the reference metal plates in the experimental testing (associated with the first two wafers W**1**, W**2**) correspond to a metal stack similar to what is shown in FIG. **9** with the two layers **902**, **908** of titanium aluminide having the same thickness. Such metal plates resulted in a final shape characterized by a negative curvature (e.g., the metal plates were cupped downward with the exterior surface **206** being convex as shown in FIG. **6**).
(43) The remaining three wafers identified in FIG. **13** (W**3**, W**4**, W**5**) include DMDs with metal plates in which the third metal layer (corresponding to a second layer of titanium aluminide) is omitted or excluded. That is, the third, fourth, and fifth wafers (W**3**, W**4**, W**5**) correspond to a metal stack similar to what is shown in FIG. **8**. Different ones of these experimental DMDs were implemented with different thicknesses for the layer **802** including 50 Å for W**3**, 100 Å for W**4**, and 150 Å for W**5**. In these experimental examples, as the thickness of the layer **802** increased, the thickness of the layer **804** correspondingly decreased (e.g., from 925 Å for W**3**, to 875 Å for W**4**, and to 825 Å for W**5**). Measuring the curvature of the metal plates in these DMDs reveal less curvature than the reference metal plates discussed above. Furthermore, the amount and nature of the curvature depends on the thickness of the layer **802**. More particularly, at thicknesses of 50 Å and 100 Å, the curvature remained negative but was closer to 0 (e.g., less curved and/or flatter) than the reference metal plates. In the examples where the thickness of the layer **802** was increased to 150 Å, the measured values for the curvature became positive (e.g., the metal plates were cupped upward with the exterior surface **206** being concave as shown in FIG. **7**). Such measurements demonstrate that it is possible to control or adjust the final shape of metal plates by adjusting the

thickness of the layer **802** of titanium aluminide.
(44) The five additional wafers identified in FIG. **14** (W**6**, W**7**, W**8**, W**9**, W**10**) include DMDs with metal plates in which the third metal layer (corresponding to a second layer of titanium aluminide) is fixed at a thickness of 100 Å, while the first metal layer (also corresponding to titanium aluminide) changes thickness between the different wafers from 150 Å (for W**5**) to 350 Å (for W**10**) in 50 Å increments. That is, in the examples, there are two layers of titanium aluminide in a similar manner to that shown in FIG. **9**. In these examples, as the thickness of the layer **902** increased, the thickness of the layer **904** (containing substantially pure aluminum) correspondingly decreased (e.g., from 725 Å (for W**5**) to 525 Å (for W**10**) in 50 Å increments). Measuring the curvature of the metal plates in each of these DMDs reveal less curvature than the reference metal plates discussed above. Furthermore, as the thickness of the first metal layer (e.g., the layer **902** of FIG. **9**) increases, the curvature changes from the most negative curvature (e.g., the plate is cupped downward) for the reference metal plates in which the layer **902** had a thickness of 100 Å) to a most positive curvature (e.g., the plate is cupped upward) for the example metal plates in which the layer **902** had a thickness of 350 Å) with the curvature being closest to 0 (e.g., closest to a flat plane) when the layer **902** had a thickness of 250 Å. As a result of the differences in curvatures based on the differences in thicknesses of the layer **902**, it is possible to select a suitable thickness that can achieve planar or substantially planar surface (or any other suitable shape or curvature) for the metal plates.
(45) Significantly, the changes in the measured values of curvature of the fifth through tenth wafers (W**6**, W**7**, W**8**, W**9**, W**10**) are incremental in a similar manner to the incremental changes to the thicknesses of the layer **902** of titanium aluminide. In other words, the relationship between changes to the thickness of the titanium aluminide and the resulting curvature of the metal plate is generally linear. As a result, it is possible to precisely tune or control the resulting shape or curvature of a metal plate by selecting the particular thickness corresponding to the desired shape. As discussed above, for micromirrors, it is generally desirable to have the curvature be as close to 0 as possible to provide a substantially planar mirror surface. Thus, in this example, the thickness of 250 Å associated with the eighth wafer (W**8**) in FIG. **14** is a good selection as the resulting curvature is close to 0. However, in other circumstances, for different MEMS devices and/or micromirrors manufactured using different parameters for the fabrication processes, a different thickness for the titanium aluminide may be more suitable. Furthermore, in some examples, a planar surface may not be needed and/or desired. For instance, a mirror surface of a micromirror that is slightly concave may be desirable to reduce the shorting margin and hinge memory lifetime, thereby providing higher yields and/or improved reliability. The generally linear relationship between titanium aluminide thickness and curvature shown in FIG. **14** establishes that any other suitable shape or curvature can be achieved with a relatively high degree of precision inasmuch as thicknesses of layers can be precisely controlled through deposition processes used to create the layers.
(46) Furthermore, because the curvature can be precisely controlled merely by adjusting the thickness of the titanium aluminide, there is no longer a need to include an air-break in the fabrication process that adds a native oxide film to the metal layers in the metal stack of the metal plate that extend into and form the post **208**. Notably, subsequent processing of the post **208** (e.g., adding the filler material **814**) will involve exposing the most recently deposited layer of material immediately beneath the filler material **814** to air such that there is likely to be an oxide layer that forms on that layer. However, in some examples, the metal stack in the micromirror structures **800**, **900**, **1000**, **1100**, **1200** disclosed herein, does not include an oxide interlayer or film between layers of metal that extend into and/or define the wall of the post **208**. Eliminating oxide interlayers within the stack in this matter serves to reduce (e.g., avoid) defectivity concerns that can arise from the presence of such oxide interlayers.
(47) The foregoing examples of the micromirror structures **800**, **900**, **1000**, **1100**, **1200** of FIGS. **8-**

12 and the various thicknesses of metal layers set forth in the experimental testing discussed above in connection with the charts in FIGS. **13** and **14** teach or suggest different features. Although each of the example micromirror structures **800**, **900**, **1000**, **1100**, **1200** of FIGS. **8-12** disclosed above have certain features, it should be understood that it is not necessary for a particular feature of one example to be used exclusively with that example. Instead, any of the features described above and/or depicted in the drawings can be combined with any of the examples, in addition to or in substitution for any of the other features of those examples. One example's features are not mutually exclusive to another example's features. Instead, the scope of this disclosure encompasses any combination of any of the features.

(48) FIG. **15** is a flowchart illustrating an example method of manufacturing a micromirror assembly with any one of the micromirror structures **800**, **900**, **1000**, **1100**, **1200** of FIGS. **8-12**. The example method of manufacture detailed in FIG. **15** will be described with reference to FIG. **16-25**, which illustrate an example micromirror assembly at various stages during the fabrication process outlined in the flowchart of FIG. **15**. Although the example method of manufacture is described with reference to the flowchart illustrated in FIG. **15** in conjunction with the example stages of fabrication represented in FIGS. **16-25**, many other methods may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be combined, divided, re-arranged, omitted, eliminated, and/or implemented in any other way. Further, additionally operations not specifically represented by the blocks in FIG. **15** may be included when implementing the example method.

(49) The example process of FIG. **15** assumes that particular design parameters for the different layers in the metal stack that defined the micromirror structure have already been selected. In some examples, the design parameters include the placement or ordering of layers in the metal stack of the micromirror structure as well as the thicknesses of the layers. As discussed above, the particular selection of design parameters enable the particular control of the stress gradient in the plate and, therefore, the particular control of the final shape of the plate. For purposes of explanation it is assumed that the selected ordering of layers is in accordance with the micromirror structure **900** of FIG. **9**. Accordingly, the blocks set forth in FIG. **15** are specified to fabricate such a micromirror structure **900**. In other examples, the process flow may be suitably adapted to fabricate any other micromirror structure disclosed herein.

(50) The example process of FIG. **15** begins at block **1502** by obtaining an underlying structure for a micromirror assembly. This stage of fabrication is represented in FIG. **16**. In some examples, as shown in FIG. **16**, the underlying structure includes a semiconductor substrate **1602** (similar to the substrate **202** of FIG. **2**), one or more electrodes **1604** (similar to any one of the electrodes **220**, **412**, **510** of FIGS. **2-5**), and a hinge **1606** (similar to any one of the hinges **214**, **408**, **508** of FIGS. **205**). Further, at this point in the fabrication process, the underlying structure also includes a sacrificial material **1608** that fills in the space surrounding the hinge **1606**, the electrodes **1604** and the substrate **1602**. The underlying structure for the micromirror assembly can be fabricated using any suitable processes now known or developed in the future. Further, the processes involved can be suitable adapted to fabricate the underlying structure with any suitable design (e.g., corresponding to any one of the micromirror assemblies **104**, **400**, **500** of FIGS. **2-5**).

(51) The stage of fabrication corresponding to block **1504** is represented in FIG. **17**. Specifically, at block **1504**, the method includes adding a sacrificial material **1702** on the underlying structure. More particularly, the sacrificial material **1702** is deposited on the exposed upper surface of the underlying sacrificial material **1608** as well as the exposed upper surface of the hinge **1606** and the exposed surfaces of the electrodes **1604**. In this examples, the sacrificial material **1702** fills the hollow interior of the electrodes **1604**. The deposition of the sacrificial material **1702** may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), spin coating, etc.).

(52) The stage of fabrication corresponding to block **1506** is represented in FIG. **18**. Specifically, at

block **1506**, the method includes creating openings **1802** in the sacrificial material **1702**. The openings **1802** may be created using any suitable process (e.g., drilling, etching, photolithography. etc.). In some examples, the openings correspond to locations where the posts to support metal plates of a micromirror structure are to be positioned.

(53) The stage of fabrication corresponding to block **1508** is represented in FIG. **19**. Specifically, at block **1508**, the method includes depositing a layer **1902** of titanium aluminide with the thickness defined by the design parameters. The deposition of the layer **1902** of titanium aluminide may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), electroplating, etc.). As shown in the illustrated example of FIG. **19**, the layer **1902** of titanium aluminide covers the exposed upper surface of the sacrificial material **1702** and also covers the wall of the opening in the sacrificial material **1702** that is to define the walls for a post. In some examples where the bottom layer of titanium aluminide is to be omitted (as in the example micromirror structure **1000** of FIG. **10**), block **1508** is omitted. That is, block **1508** is optional.

(54) The stage of fabrication corresponding to block **1510** is represented in FIG. **20**. Specifically, at block **1510**, the method includes depositing a layer **2002** of aluminum with the thickness defined by the design parameters. In this example, the layer **2002** of aluminum is deposited directly onto the previously deposited layer **1902** of titanium aluminide. The deposition of the layer **2002** of aluminum may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), electroplating, etc.).

(55) The stage of fabrication corresponding to block **1512** is represented in FIG. **21**. Specifically, at block **1512**, the method includes depositing a layer **2102** of titanium aluminide with the thickness defined by the design parameters. In this example, the layer **2102** of titanium aluminide is deposited directly onto the previously deposited layer **2002** of aluminum. The deposition of the layer **2102** of titanium aluminide may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), electroplating, etc.). In some examples where the upper layer of titanium aluminide is to be omitted (as in the example micromirror structure **800** of FIG. **8**), block **1512** is omitted. That is, block **1512** is optional. In examples where the upper layer of titanium aluminide is to be deposited on particular regions rather than across the entire underlying surface (as in the example micromirror structure **1200** of FIG. **12**), a photoresist may be deposited and patterned before the titanium aluminide is deposited.

(56) The stage of fabrication corresponding to block **1514** is represented in FIG. **22**. Specifically, at block **1514**, the method includes depositing a filler material **2202** in the post **2204** defined by the metal layers **1902**, **2002**, **2102** within the opening of the sacrificial material **1702**. The deposition of the layer **2302** of aluminum may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etc.). In some examples, after the filler material **2202** is deposited a planarization process may be implemented to make the exposed surface of the filler material **2202** is even with the exposed surface of the metal layer **2102**. If the metal layer **2101** is to extend across the top of the filler material **2202** (as in the example micromirror structure **1100** of FIG. **11**), block **1514** may be implemented before block **1512**. Block **1514** is optional. Accordingly, in some examples, block **1514** is omitted.

(57) The stage of fabrication corresponding to block **1516** is represented in FIG. **23**. Specifically, at block **1516**, the method includes depositing a layer **2302** of aluminum with the thickness defined by the design parameters. In this example, the layer **2302** of aluminum is deposited directly onto the previously deposited layer **2102** of titanium aluminide and across the filler material **2202**. The deposition of the layer **2302** of aluminum may be accomplished through any suitable process (e.g., atomic-layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), electroplating, etc.).

(58) The stage of fabrication corresponding to block **1518** is represented in FIG. **24**. Specifically, at block **1518**, the method includes patterning and etching the metal plate **2402** for the micromirror assembly. That is, up to this point, the stack of metals correspond to a sheet of metal that is divided (at block **1518**) into individual metal plates for individual micromirrors. The patterning and etching may be accomplished through any suitable process (e.g., lithographic techniques).
(59) The stage of fabrication corresponding to block **1520** is represented in FIG. **24**. Specifically, at block **1520**, the method includes removing the sacrificial materials **1608**, **1702** to release the micromirror structure along with the rest of the micromirror assembly. The removal of the sacrificial material may be accomplished using any suitable technique (e.g., a wet etch process). Thereafter, the example method of FIG. **15** ends.
(60) The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.
(61) As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.
(62) From the foregoing, it will be appreciated that example systems, methods, apparatus, and articles of manufacture have been disclosed that enable the stress gradient in components of MEMS devices to be modified in a controlled manner so as to control the final shape of the components. More particularly, the stress gradient of aluminum components can be precisely controlled by including one or more layers of titanium aluminide in the components and adjusting the thickness of the layers to achieve any suitable final shape for the component. While examples described above and detailed in the figures correspond to mirror surfaces used in micromirror structures of DMDs, teachings disclosed herein are not limited to such applications but can be suitably adapted to control or modify stress gradients in any aluminum-based MEMS structures, components, or elements (e.g., plates, bridges, cantilevers, crab arms, etc.) used in any other type of MEMS devices (e.g., microbolometers, microphones, accelerometers, contact switches, etc.) to control the final shape of such MEMS structures.
(63) Further examples and combinations thereof include the following:
(64) Example 1 includes a microelectromechanical system (MEMS) device comprising a substrate, and a MEMS structure supported by the substrate, the MEMS structure comprising a first layer of a first material comprising a titanium aluminum alloy, and an aluminum layer on the first layer.
(65) Example 2 includes the MEMS device of example 1, wherein the MEMS structure comprises a

second layer of a second material, the aluminum layer of the MEMS structure between the first and second layers of the MEMS structure, the second material the same as the first material.
(66) Example 3 includes the MEMS device of example 2, wherein the aluminum layer is a first aluminum layer, the MEMS structure comprises a second aluminum layer, the second layer of the MEMS structure between the first and second aluminum layers of the MEMS structure.
(67) Example 4 includes the MEMS device of example 3, wherein the second aluminum layer of the MEMS structure forms an exterior surface of the MEMS structure.
(68) Example 5 includes the MEMS device of example 1, wherein the aluminum layer is a first aluminum layer, the MEMS structure comprises a second aluminum layer, the first aluminum layer of the MEMS structure between the first layer and the second aluminum layer of the MEMS structure.
(69) Example 6 includes the MEMS device of example 1, further comprising a hinge supported by the substrate, the hinge configured to rotate the MEMS structure, and a post coupling the MEMS structure to the hinge, the first material extending across a top of the post.
(70) Example 7 includes the MEMS device of example 6, further comprising a filler material within the post beneath the first material.
(71) Example 8 includes the MEMS device of example 1, further comprising a hinge supported by the substrate, the hinge configured to rotate the MEMS structure, and a post coupling the MEMS structure to the hinge, the first material extending along a wall of the post.
(72) Example 9 includes the MEMS device of example 1, wherein the first material is titanium aluminide.
(73) Example 10 includes the MEMS device of example 1, wherein the MEMS structure is a mirror, and the MEMS device is a digital micromirror device.
(74) Example 11 includes the MEMS device of example 1, wherein the first layer extends across a first area, and the aluminum layer extends across a second area, the first area corresponding to less than all the second area.
(75) Example 12 includes a microelectromechanical (MEMS) device comprising a substrate, a hinge, a pillar between the hinge and the substrate, and a micromirror coupled to the hinge via a post, the micromirror comprising a first layer of metal, the first layer comprising aluminum, and a second layer of metal below the first layer, the second layer comprising a titanium aluminum alloy.
(76) Example 13 includes the MEMS device of example 12, wherein the micromirror comprises a third layer of metal between the first layer and the second layer, the third layer comprising aluminum.
(77) Example 14 includes the MEMS device of example 13, wherein the micromirror comprises a fourth layer of metal between the first and third layers, the fourth layer comprising a titanium aluminum alloy.
(78) Example 15 includes the MEMS device of example 14, wherein the second layer is thicker than the third layer, and the third layer is thicker than the fourth layer.
(79) Example 16 includes the MEMS device of example 12, wherein the first layer defines a first exposed surface of the micromirror, and the second layer of metal defines a second exposed surface of the micromirror, the first and second exposed surfaces facing in opposite directions.
(80) Example 17 includes the MEMS device of example 12, wherein the micromirror comprises a third layer of metal below the second layer, the third layer comprising aluminum.
(81) Example 18 includes a method comprising obtaining a substrate, depositing a first layer of metal over the substrate, the first layer of metal comprising a titanium aluminum alloy, and depositing a second layer of metal over the first layer of metal, the second layer of metal being an aluminum layer.
(82) Example 19 includes the method of example 18, further comprising depositing a third layer of metal over the second layer of metal, the third layer of metal comprising a titanium aluminum alloy.

(83) Example 20 includes the method of example 18, further comprising depositing a third layer of metal over the second layer of metal, the second layer of metal being an aluminum layer.
(84) The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, methods, apparatus, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, methods, apparatus, and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. A microelectromechanical system (MEMS) device comprising: a substrate; a post supported by the substrate; and a MEMS structure supported by the post, the MEMS structure comprising: a first layer of a first material comprising a titanium aluminum alloy; and an aluminum layer over the first layer, the aluminum layer over the post.
2. The MEMS device of claim 1, wherein the MEMS structure comprises a second layer of a second material, the aluminum layer of the MEMS structure between the first and second layers of the MEMS structure, the second material the same as the first material.
3. The MEMS device of claim 2, wherein the aluminum layer is a first aluminum layer, the MEMS structure comprises a second aluminum layer, the second layer of the MEMS structure between the first and second aluminum layers of the MEMS structure.
4. The MEMS device of claim 3, wherein the second aluminum layer of the MEMS structure forms an exterior surface of the MEMS structure.
5. The MEMS device of claim 1, wherein the aluminum layer is a first aluminum layer, the MEMS structure comprises a second aluminum layer, the first aluminum layer of the MEMS structure between the first layer and the second aluminum layer of the MEMS structure.
6. The MEMS device of claim 1, further comprising: a hinge supported by the substrate, the hinge configured to rotate the MEMS structure; and the post coupling the MEMS structure to the hinge, the first material extending across a top of the post.
7. The MEMS device of claim 6, further comprising a filler material within the post beneath the first material.
8. The MEMS device of claim 1, further comprising: a hinge supported by the substrate, the hinge configured to rotate the MEMS structure; and the post coupling the MEMS structure to the hinge, the first material extending along a wall of the post.
9. The MEMS device of claim 1, wherein the first material is titanium aluminide.
10. The MEMS device of claim 1, wherein the MEMS structure is a mirror, and the MEMS device is a digital micromirror device.
11. The MEMS device of claim 1, wherein the first layer extends across a first area, and the aluminum layer extends across a second area, the first area corresponding to less than all the second area.
12. A microelectromechanical (MEMS) device comprising: a substrate; a hinge; a pillar between the hinge and the substrate; and a micromirror coupled to the hinge via a post, the micromirror comprising: a first layer of metal, the first layer comprising aluminum; and a second layer of metal between the first layer and the hinge, the second layer comprising a titanium aluminum alloy.
13. The MEMS device of claim 12, wherein the micromirror comprises a third layer of metal between the first layer and the second layer, the third layer comprising aluminum.
14. The MEMS device of claim 13, wherein the micromirror comprises a fourth layer of metal between the first and third layers, the fourth layer comprising a titanium aluminum alloy.
15. The MEMS device of claim 14, wherein the second layer is thicker than the third layer, and the third layer is thicker than the fourth layer.
16. The MEMS device of claim 12, wherein the first layer defines a first exposed surface of the

micromirror, and the second layer of metal defines a second exposed surface of the micromirror, the first and second exposed surfaces facing in opposite directions.
17. The MEMS device of claim 12, wherein the micromirror comprises a third layer of metal below the second layer, the third layer comprising aluminum.
18. A microelectromechanical system (MEMS) device comprising: a substrate; a post supported by the substrate; and a MEMS structure supported by the post, the MEMS structure comprising: a first layer of a first material comprising a titanium aluminum alloy, the first layer forming at least a portion of the post; and a second layer over the first layer, the second layer being an aluminum layer.
19. The MEMS device of claim 18, wherein the second layer forms at least a portion of the post.
20. The MEMS device of claim 18, wherein the second layer extends over the post.